Number Theory

Question

Find multiplicative inverse of 356 mod 45

Solution :

> $u = u_o - q.u_1$

repeat until r = 0

q	a	m	r	u_o	u_1	u
7	356	45	41	1	0	1
10	41	4	1	1	-1	11
4	4	1	0	-1	11	

GCD = 1; u_1 = Multiplicative index = 11

356 X 11 mod 45 = 1

Congruence Class

- A congruence class or residual class is a set of integers that are equivalent to each other modulo n

> a = (congruency symbol) b mod n (a-b) mod n = 0 a mod n = b mod n

- a and b gives same remainder when divided by n

Question

12 = 18 (mod 15)

Solution

12 mod 15 = 12 18 mod 15 = 3 This is invalid

Question

-3 = -18 (mod 15) **Solution** -3 mod 15 = 0 ; -18 mod 15 = 12 valid

Question

12 = -3(mod 15) **Solution** 12 mod 15 = 12 ;-3 mod 15 = -3 valid

Question 7 = 19(mod 11) **Solution** 7 mod 11 = 0; 19 mod 11 = 8 invalid

Question 4 = -19(mod11) **Solution** 4 mod 11 = 0; -19 mod 11 = -8 invalid

Question 4 = -18 (mod 11) **Solution** 4 mod 11 = 0; -18 mod 11 = -7 valid

Question 4^{-1} = 2 mod 7 **Solution** 4^{-1}mod 7 = ; 2 mod 7 = 0 1/4 = 2(mod 7) valid

Question Find if MI exists 2^{-1} mod 10

Solution

> GCD (a,n) = 1 GCD of 2 and 10 is 2 so not valid

Question 2^{-1} mod 11 **Solution** GCD = 1 so GCD exists

Question 3^{-1} mod 10 **Solution** GCD exists

Question Find MI of 9 mod 10 **Solution**

q	a	m	r	u_o	u_1	u

Question Find MI of 23 in z_{100} **Solution** u_1 = -13 -13 mod 100 100 + (-13) 87

Question 53 in z_{54} **Solution**

Multiplicative Inverse Pair

Two numbers who are MI of each other

Question Find MI pair for Z_{10} and Z_{11} **Solution** MI of 1 mod 10 = 1 MI of 2 mod 10 = 2 MI of 3 mod 10 = 3 MI of 4 mod 10 = 4 MI of 5 mod 10 = 5 MI of 6 mod 10 = 6 MI of 7 mod 10 = 7 MI of 8 mod 10 = 8 MI of 9 mod 10 = 9 MI of 10 mod 10 = 0

Z_{10} = {(1,1), (3,7),(9,9),(7,3)}

Z_{11} = 10 pairs

Euler's Totient Function

> Z^*_m = {0,...,m-1}

- Z_m is the set of numbers that are relatively prime or co-prime with m in Z_m
- Z^*_m can be found out by knowing if GCD of m and other no. should be 1
- Z^*_5 = {1,2,3,4}
- phi (m) - ETF is denoted this way which gives the no. of elements in Z^*_m
- phi(m) = $|Z^*_m|$ (cardinality)

- Z^*_{14} = {1,3,5,9,11,13}, phi(Z14) = 6

Conditions of phi (m)

- phi (p) = p-1 ; p=prime no.
- phi (p) = phi (mXn) = phi (m) X phi (n) ; m,n are co-prime
- phi (m) = phi (p^e) = p^e - p^{e-1}

Example

- phi (16) = phi (2^4) = 2^4 - 2^3 = 8
- phi(30) = 10
- phi(4) =
- phi(240) = 4

Questions

1. phi (27) = 3^3 = 3^3 - 3^2 = 18
2. phi(630) = phi(10) X phi(63) = phi(5X2) X phi(9X7) = 1 X 4 X phi(3^2) X 6 = 24 X (3^2- 3^1) = 24 X 6 = 144
3. phi(49) = 7^2 = 7^2 - 7^1 = 42
4. phi(1323) = 3^3 X 7^2
5. phi(287375) = phi(6^3) X phi(11) = 198000

Fast Exponentiation Algorithm

> A^B mod C

1. Convert B to binary value
2. Compute **A^i mod C** using square method. Stop when **i < = B**

A^1 mod C = x A^2 mod C = y A^4 mod C A^8 mod C ....

3. Find A^B mod C using selective multiplication method

B = 5 = > 101 = A^5 mod C A^{101} mod C A^{4+1} mod C

Question 11^{13} mod 53 **solution** 13 - 1101 - 8+4+1 11^1 mod 53 = 11 11^2 mod 53 = (11 mod 53 X 11 mod 53) mod 53 = (11 X 11) mod 53 = 15 11^4 mod 53 = (11^2 mod 53 X 11^2 mod 53) mod 53 = (15 X 15) mod 53= 13 11^8 mod 53 = 10 11^{13} mod 53 = 11^{8+4+1} mod 53 = (11^8 mod 53 X 11^4 mod 53 X 11^1 mod 53) mod 53 = (10 X 13 X 11) = 52

Question 5 ^{117} mod 19 **solution** 117 - 1110101 - 5

5 mod 53 = 5 5^2 mod 19 = 5 5^4 mod 19 = 25^2 mod 19

Questions 15^{19} mod 37 23^{971} mod 503

Fermat's Little Theorem

a is an integer, p is a prime no. If p dooes not divide a then a $^{p-1}$ $\equiv$ 1 (mod p)

2^{566} $\equiv$ mod (567). Is it true?

p = 567, a = 2 567 is not prime. It is not true

Find 3^{-1} (mod 31)

3^{30} = 1 (mod 31) [Divide by 3] 3^{29} = 1/3 (mod 31) 1/3 = 3^{29} (mod 31) 3^{-1}=3^{29} (mod 31) 3^{-16} * 3^{8} 3^{4} 3(mod 31)

Euler's Theorem

Let **a** and **n** be coprime numbers. phi(n) be the Euler's function.

> a$^{phi(n)}$ $\equiv$ 1(mod n)

Compute 6^{24} mod 35 a = 6; n = 35; a and n are coprime phi(n) = phi(35) = phi(7 X 5) = 6 X 4 = 24 6^{24} $\equiv$ 1 mod 35

20^{62} mod 77 phi(n) = phi(77) = phi(11 X 7) = 10 X 6 = 60 20^{60} mod 77 20^{60} mod 77 X 20^{2} mod 77 1 X 400 mod 77 = 15

Compute 71^{-1} mod 100 using Euler's theorem

phi(100) = phi(25) X phi(4) = 40 71^{40} $\equiv$ 1 mod(100) 71^{-1} mod 100 = 1/71 mod 100 71^{40} mod(100) / 71 mod 100 71^{40-1} mod(100) 71^{39} mod(100)

Use fast exponentiation to compute 71^{39} mod(100)

8^{-1} mod(77)

phi(77) = 60 8^{60} $\equiv$ 1 mod(77) 8^{-1} mod (77) = 1/8 mod 77 8^{60} mod (77) / 8 mod 77 8^{60-1} mod (77) 8^{59} mod (77) 29

Primitive Element

p is a prime no. and Z_p = {0,1,2,3,.....,p-1}

> phi(p) = p-1 and a^{p-1} $\equiv$ 1 mod (p)

Then a is said to be a primitive element in Z_p

Public Key Cryptography

- Public key / Two key / Asymmetric key
- Uses 2 keys :
- Public - It can be known to anyone, is used to encrypt messages and verify signatures.
- Private - It is only known to the recipient , used to decrypt messages and create signatures.

Public Key Cryptosystems

> $Z = E(KU_b, E(KR_a,X))$ $X = D(KU_a, D(KR_b,Z))$

- Public Key Cryptosystem (PKC) must be exercised 4 times
- It is used in digital signatures, key exchange, encryption/decryption

Algorithm	Encryption/Decryption	Digital Signature	Key Exchange
RSA	Yes	Y	Y
Eliptic Curve	Y	Y	Y
Diffie-Hellman	N	N	Y
Digital Signature System(DSS)	N	Y	N

Requirements for PKC

- It is computationally easy for a party B to generate a pair
- For sender A

> $C = E(KU_b, M)$

- For reciever B

> $M = D(KR_b, C) = D [KR_b, E (KU_b, M)]$

- KU_b - Public key, KR_b - Private Key

> $M = D[KU_b, E(KR_b,M)] = D[KR_b, E(KU_b,M)]$

- **Trap door one way function** - easy to calculate in one direction but is computationally hard to do the inverse unless certain additional information is known.
- Example: $Y = f(X)$ is easy but $X = f^{-1}(Y)$ is infeasible

Misconceptions about PKC

- It replaces symmetric crypto - PKC complements private key crypto
- PKC is secure - no evidence found
- Key distribution is trivial - making something public is not trivial but easy

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RSA (Rivest, Shamir, Adelman) Algorithm

- It is the most widely used public key cryptography algorithm in the world.
- It can be used for both public key encryption and digital signatures.

- It makes use of prime numbers
- Integers used by RSA have to be large in size
- There are two sets of keys: private and public key

Key Generation

Select p, q	p and q both prime
Calculate $n = p \times q$	
Calculate $\phi(n) = (p-1)(q-1)$	
Select integer e	$\gcd(\phi(n), e) = 1;\ 1 < e < \phi(n)$
Calculate d	$d \equiv e^{-1} \bmod \phi(n)$
Public key	$KU = \{e, n\}$
Private key	$KR = \{d, n\}$

Encryption

Plaintext	$M < n$
Ciphertext	$C = M^e \pmod n$

Decryption

Ciphertext	C
Plaintext	$M = C^d \pmod n$

Example Problem

P = 3, Q = 11 M = 14 C =?; C = 13 M = ?

n = 33 phi(n) = 2 X 10 = 20 Set of co-prime - Z^*_n = {1,2,4,5,7,8,10,13,14,16,17,19,20..} e = 7 (gcd should be 1 from above set of numbers) d $\equiv$ e^{-1} mod phi(n) d = 7^{-1} mod phi(20) d = 3

Public Key - KU = {7,33}, Private Key - KR = {3,33} Encryption: Plain Text - M<n - 14<33 Cipher Text - C = 14^7 mod 33 C = 20

Decryption: Cipher Text - 13 Plain Text - M = 13^3 mod 33 M = 19

Digital Signatures using RSA

> $D_{K\ Pvt}$ ($E_{K\ Pub}$ (M) = M)

Text is encrypted using K_{Pvt} and decrypted with $_{K\ Pub}$

Discrete Logarithm

> g^x $\equiv$ h (mod p)

g is the primitive element in Z_p and x is the Discrete Logarithm of h with base g mod p

x = $dlog_{g,p}$ (h)

Example

$dlog_{g,p}$ (1) = 0

ElGamel Encryption Algorithm

- It is a public key cryptosystem which uses asymmetric key encryption
- It is based on difficulty of finding Discrete Logarithm
- It is the second category of PKC
- C1 carries the random exponent info required to compute the shared secret
- C2 contains the encryoted message

• **Key Generation**	
	Select a large prime as a q Select x *to* be a member of the group **G** = < Zq^*, X >, x must be "$1 \le x \le q-1$" Select g *to* be a primitive root (generator) in the group **G** = < Zq^*, X > $y = g^{x} \bmod q$ Public key ← (g, y, q) *Private key* ← x
• **Encryption**	
	Select a random integer r in the group **G** = < Zq^*, X >, r must be "$1 \le r \le q-1$" $C_1 = g^{r} \bmod q$ $C_2 = (p \,.\, y^{r}) \bmod q$ // p is the plaintext
• **Decryption**	
	$P = [\, C_2 (C_1{}^{x})^{-1} \,] \bmod q$

C2=M·y^rmod q

Example

Encrpyt message 10 using public key 9 and primitive root as 11 and a prime no. 23 also a random no. 3

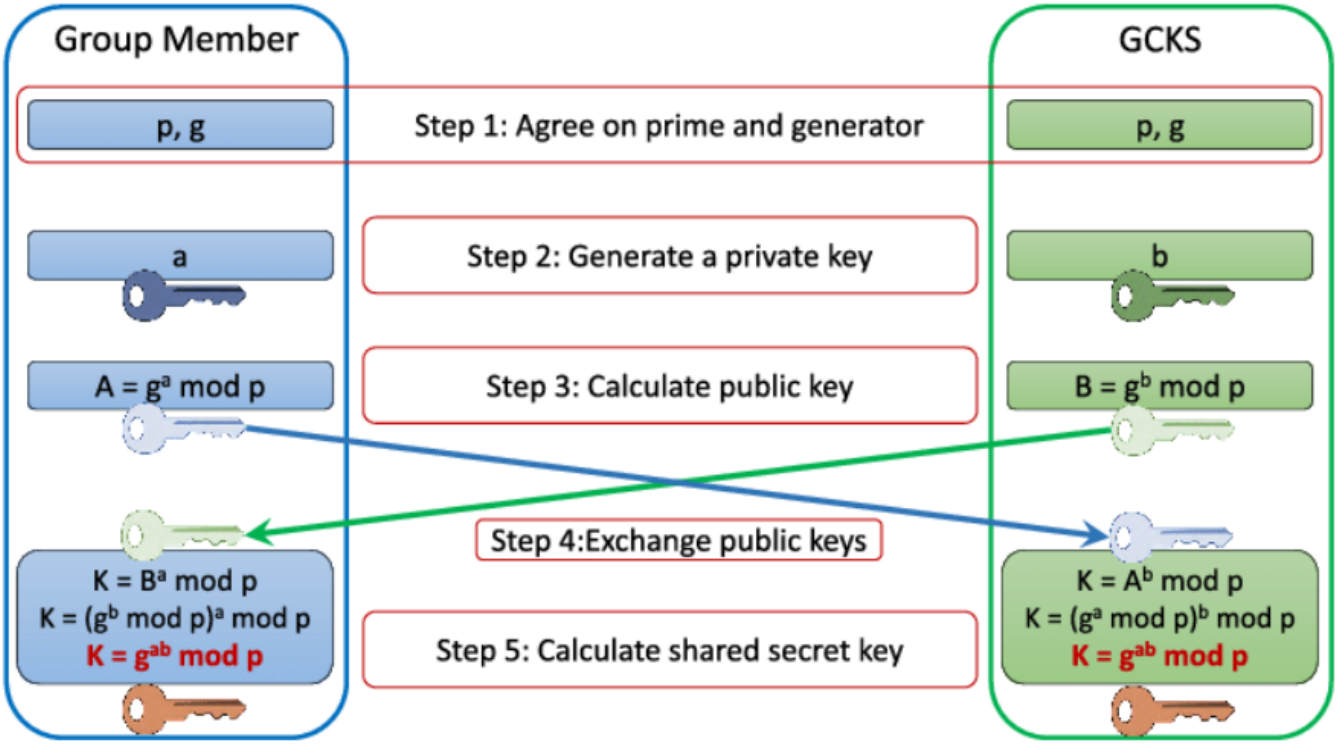
q = 23, p=10, y= 9, r=3, G=11 C1 = 11^{3} mod 23 = 20 C2 = (10.9^{3}) mod 23 = 22 P = [23 $(20^{6})^{-1}$] mod 23 x = 6 (trial and error) first 20^{6} mod 23 = 16 22 X 16^{-1} mod 23 = (22 X 16^{-1} mod 23) mod 23 (22 X 13) mod 23 = 10

Question q=23,g=7, x=9, M=20, r=3

y = 7^{9} mod 23 = 15 C1 = 7^{3} mod 23 = 21 C2 = 20 (y^{3} mod 23) = 18 p=[18$(21^{9})^{-1}$] mod 23 =

Diffie Hellman(midsem)

- Diffie Hellman algorithm is used for key exchange mechanism to verify if a right key is shared by the right person.
- If a third party has identified a common secret, it would be still difficult to determine the private key



Question P = 13, g = 6, a = 5, b = 4

A = 6^5 mod 13 = 2 B = 6^4 mod 13 = 9 K_a = 9^5 mod 13 = 3 = 6^{20} mod 13 = K_b = 2^4 mod 13 = 3 = 6^{20} mod 13 =